26 **D**

Calculus method:

For a solenoid with a current *I*, and *I* varies with time as given:

$$B \propto I = I_0 \sin^2 \omega t$$

Hence using Faradays law to find the emf in the smaller coil:

$$E = A\frac{dB}{dt} = kI_0 \sin 2\omega t = E_0 \sin \omega' t$$

The result is a graph with twice the frequency of the graph given.

Reasoning method:
For a solenoid with a current *I*, $B \propto I$
Consider the **first half period**: the current increases then decreases, so the magnetic flux experienced by the smaller coil also increases and decreases,
Applying Faraday's Law and Lenz's Law, the emf induced will be in opposite directions during the increase compared to the decrease.
Thus for one complete period as shown in the graph, the emf induced will be positive, then negative then positive then negative again.

27 B